THE DRIFT PRIMER

The Case for Native Digital Chaos in the Post-Quantum Era

Prepared For: Strategic Partners & Technical Review

Date: December 2025

Status: TRL-5 FPGA-Verified (>24B Cycles, HW/SW lockstep)

IP Portfolio: 63 Provisional Patent Applications Filed

Contents

1 The Entropy Crisis

The industry is currently trapped between two inadequate paradigms for generating entropy. **Analog TRNGs** rely on physical phenomena like thermal noise or oscillator jitter. While statistically random, they suffer from inherent physical latency blocking (waiting for entropy to accumulate) and environmental bias (temperature drift), rendering them unreliable for high-speed, precision control loops.

Conversely, standard **Digital CSPRNGs** (like AES-CTR) provide robust security but impose a heavy computational tax. They rely on complex S-Box substitutions and matrix multiplications that require over 3,500 logic gates and multiple clock cycles to initialize. This architecture generates significant dynamic power heat—a "Thermal Tax"—that is prohibitive for cryogenic environments, bio-implants, and energy-starved edge devices.

There is currently no primitive that delivers high-quality conditioned entropy with the **zero-latency speed** of a simple logic gate and the **near-zero thermal profile** required by next-generation hardware.

The Drift Thesis

The industry needs Native Digital Chaos. We generate high-quality conditioned entropy using the inherent physics of digital logic—Shift and Add—without the overhead of multipliers or analog circuitry.

2 Discrete Arithmetic Dynamics (DAD)

The core innovation of Drift Systems is the rejection of "simulated" randomness in favor of **Native Digital Chaos**. We utilize the inherent non-linear properties of binary addition to generate entropy.

2.1 The Core Mechanism

The Drift Engine operates on a 128-bit (or 256-bit) state register. It generates a stream of non-repeating, high-entropy numbers by cycling through three distinct atomic operations in a single clock cycle:

1. **THE PUSH (Expansion):** The state S is subjected to an affine transformation $(qS + d)$. The multiplication by an odd integer (e.g., $q = 3$) creates a **Carry Chain Avalanche**.
2. **THE SQUEEZE (Drift):** We prevent short cycles via Forced Drift. By restricting the division step, we guarantee the internal trajectory wants to grow to infinity.

3. **THE FOLD (Mixing):** Since infinite growth is impossible on a finite chip, we Fold the overflow back into the lower bits, creating a non-linear trapdoor.

2.2 The Unified Core (Stream + Hash)

Unlike legacy chips that require separate blocks for Encryption (AES) and Hashing (SHA), the Drift Core is **Reconfigurable**.

- **Stream Mode:** The core generates infinite entropy for encryption.
- **Sponge Mode (Hash):** The core absorbs external data (via XOR injection) and mixes it using the arithmetic carry chain.
- **Benefit:** A single $\sim$580-LUT circuit (Tang Primer 20K, hardware-validated) performs both functions, reducing silicon area by 50%.

3 The Trust Architecture: Provable Math

For a new cryptographic primitive, trust is paramount. Users must know that the generator does not hide ”Dark Corners” (unreachable states) or ”Backdoors” (secret cycles). We address this via two fundamental mathematical proofs.

3.1 1. Provable Routing Completeness (Dynamic Covering Systems)

We utilize a **Dynamic Covering System** (Erdős Covering). The front-end routing logic dispatches every integer state into exactly one of a small set of transition gates (T1/T2/T3), a property formally verified in Lean 4 (`Completeness.lean`; no `sorry`, no axioms beyond Mathlib). This eliminates ”Orphan States” in the dispatching logic, ruling out shadow triggers in the front-end.

Scope note. Routing completeness is distinct from per-orbit trajectory coverage of the iterated recurrence. The production `driftStep` is provably 2-to-1 (`Noninjectivity.lean`), so each orbit is confined to a recurrent set R with $|R| \leq 2^{W-1}$ (`RecurrentSet.lean`). The strict measure-preserving ”ergodicity” reading is therefore false on $\mathrm{Fin}(2^W)$; what is true is structural routing completeness on the front-end plus a cyclic permutation on R.

3.2 2. Forward Indistinguishability (Generalized Collatz Dynamics)

The arithmetic step is built on the Generalized Collatz family ($qS + d$ with q, d parameters; production instantiation $q = 3$, $d = 1$, output via integer divide by 2). Predicting the state at $t + k$ without running k forward steps is, for the parameters we use, an open computational problem. This motivates the recurrence's use as a candidate **Verifiable Delay Function (VDF)** primitive; formal cryptographic-strength claims remain gated on independent peer review (`driftsystems.io/docs/entropy_barriers.html`).

4 Metrics & Validation (TRL-5 FPGA-Verified)

4.1 1. The Silicon Footprint (2,828 GE ASIC / ∼580 LUT FPGA)

ASIC (SKY130HD, area-opt, 13.56 MHz): 2,828 NAND2-equivalent gates ($17{,}690.72\,\mu$m^2, 0.0177 mm^2). Measured via Yosys + OpenLane on the SkyWater 130nm open-source PDK.
FPGA (Tang Primer 20K, Gowin GW2A-18): ∼580 LUT (hardware-validated, bit-exact lockstep against the cycle-accurate software model).
Industry Reference: AES-128 hardware $\approx 3{,}000$–$3{,}500$ GE; Ascon $\approx 2{,}300$–$3{,}000$ GE; PRESENT $\approx 1{,}570$ GE; SIMON 32/64 $\approx 1{,}000$ GE.
Positioning: Sub-AES, NIST-LWC class. Passes the NIST Lightweight Cryptography envelope ($< 3{,}000$ GE) and the EPC Gen2 chip budget ($< 2{,}000$ GE is not met, but the IoT lightweight envelope is). Does not fit the passive-RFID smart-label envelope ($< 1{,}000$ GE) at this footprint — the smart-label narrative is not yet available; the IoT lightweight narrative is.

4.2 2. Security Variant: Optional Zero-Trap Detector

For security-positioned deployments, the RTL exposes a synthesis parameter `ENABLE_ZERO_TRAP_DETECTOR` that monitors the arithmetic output and reroutes to a known-safe seed (`128'h1`) whenever the next state would be `0`. This eliminates the fixed point at zero as a structural concern — including the sole non-trivial predecessor $s^* = (2^W-1)/3$ — in one parameter-independent circuit. Overhead is roughly 150 LUT ($\approx 0.75\%$ of the Primer 20K design) and one extra LUT level on the critical path. Default is *on* for security and cryptographic variants, *off* for cost-optimized IoT variants where pure-math equivalence to the canonical recurrence is required. See `driftsystems.io/docs/compute.html#proof` §3.12.

4.3 3. Physical Validation (Soak Test 2025)

On December 4, 2025, the Drift Core reached TRL-5 (FPGA-verified) via a continuous hardware soak test.

- **Stability:** $> 24,000,000,000$ (24 Billion) cycles with **0.00% Divergence**.
- **Resilience:** Validated ”Self-Healing” capability. During a simulated ”Signal Blackout” (Host LOS), the core continued blind operation for 115 million cycles. Upon reconnection, the software shadow model automatically re-synchronized in < 100ms.

5 Strategic Applications

5.1 Defense & Aerospace: The ”Anti-Jamming” Shield

Drift-Hop™: Tactical radios hop frequencies in 1 clock cycle (Zero Latency), faster than enemy jammers can react (”Look-through” capability).

5.2 Web3 & Infrastructure: The ”Trustless” Oracle

Drift-Hash: Because the Conway Core is ”Undecidable,” it acts as a hardware-based Verifiable Delay Function (VDF). It allows decentralized networks (DePIN) to verify ”Proof of Useful Work” without relying on wasteful energy consumption (Proof of Work).

5.3 Medical & Bio-Implants

Bio-Drift™: Zero-Multiplier architecture reduces dynamic power draw to $< 5\mu W$, enabling secure authentication for pacemakers without heating tissue.

5.4 4. Logistics: ”The Internet of Disposables”

Drift-Tag™: A printed electronics label that changes its digital identity every time it is scanned, making supply chain cloning mathematically impossible.

5.5 5. Web3 & Data Integrity: The ”Trustless” Oracle

Drift-Sponge: The Unified Core validates data integrity (Hashing) and proves computational work (VDF) for blockchain networks.

- **Validation:** In TRL-5 FPGA testing, the core demonstrated a ∼**50% Avalanche Effect**, with a single input bit-flip propagating to roughly half of output bits within 20 cycles. This enables ”Hardware Hashing” on devices too small for SHA-256.

6 The Vision: Security as Physics

Drift Systems Inc. proposes a new paradigm: **Security as Physics.** By embedding the Drift Core into the fundamental logic gates of the semiconductor, we transform security from a ”Computational Task” into a ”Physical Property” of the chip.

Join the Pilot Program

Drift Systems Inc. is currently selecting strategic partners for early access to the Drift Core IP (Verilog/SystemC) and the Drift-Mark AI SDK.

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